

Scalability & Future Plan

Date	29 September 2026 – 2 October 2026
Team ID	6ab4fddea238dc999eb665e9
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how PocketSmart AI can be scaled to support larger user bases, increased recommendation workloads, stronger data storage and additional planning capabilities in the future. It also captures the roadmap for evolving the current student/demo solution.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	SQLite is suitable for the current student/demo deployment but is not intended for high-volume concurrent production workloads.	A larger user base could create database contention and make scaling across multiple instances difficult.	High
2	External shopping/service links are safe search/demo links rather than verified real-time inventory integrations.	Users do not receive guaranteed current stock, availability or live market pricing.	Medium
3	Gemini response quality and availability depend on the configured model, API key, account limits and service availability.	AI generation can vary in latency or temporarily fall back to demo recommendations.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	FastAPI + SQLite student/demo deployment.	Move to managed PostgreSQL and horizontally scalable application instances behind a production load balancer.
2	Data Storage	SQLite stores users, sessions and recommendation history.	Use managed relational storage, backups and appropriate indexing; separate large media storage from the database.
3	Performance	Gemini calls and recommendation generation are handled during user requests.	Add caching where appropriate, asynchronous/background processing for long tasks, connection pooling and performance monitoring.
4	Security	Hashed passwords, JWT-backed sessions and environment-based secrets are used.	Add centralized secret management, stronger production session controls, rate limiting, audit logging and security monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Production-ready data and deployment foundation	Near Term	Improved reliability, maintainability and readiness for more users.
Phase 2	Real-time product/service integrations and verified pricing/availability	Short Term	More useful recommendations based on current external information.
Phase 3	Advanced personalization and multimodal recommendation improvements	Medium Term	More context-aware recommendations using user preferences and visual inputs.
Phase 4	Managed infrastructure, analytics and broader planner expansion	Long Term	Greater scalability, stronger monitoring and additional planning domains.